CHAPTER 2

Literature Review

This chapter presents a comprehensive review of existing research, academic studies, and commercial implementations in the domain of smart retail, cashier-less checkout systems, and self-scanning technologies. The review aims to identify the current state of the art, analyze the strengths and limitations of prior approaches, and establish the research gap that the SCAN_GO project addresses. By examining systems from North America, Europe, and Asia, as well as relevant academic literature, this chapter provides the theoretical and practical foundation upon which SCAN_GO is designed and developed.

2.1 Introduction to Smart Retail and Cashier-less Technology

The concept of smart retail has emerged as one of the most transformative forces reshaping the global retail industry. Driven by rapid advances in computer vision, artificial intelligence (AI), the Internet of Things (IoT), and mobile computing, smart retail systems aim to create seamless, efficient, and personalized shopping experiences that reduce or eliminate reliance on traditional cashier-based workflows (Pantano et al., 2018).

Traditional supermarket checkout processes have remained largely unchanged for decades. Customers gather items, proceed to a checkout lane, and wait for a cashier to scan each product individually before completing payment. This model is increasingly inadequate in the face of growing consumer expectations for speed, convenience, and digital integration (Grewal et al., 2017).

The literature on smart retail can be broadly categorized into three streams: (1) infrastructure-heavy autonomous systems relying on overhead cameras and deep learning; (2) RFID-based item tracking systems that attach electronic tags to individual products; and (3) basket-integrated self-scanning systems where customers scan product barcodes using a hardware scanner mounted directly on the shopping basket. SCAN_GO belongs primarily to the third category, leveraging an ESP32 microcontroller, a dedicated barcode scanner, a weight sensor, and an Android tablet mounted on the shopping basket to create a cost-effective solution without significant store-wide infrastructure investment.

2.2 Amazon Go — The Pioneer of Cashier-less Retail (USA)

Amazon Go is widely regarded as the most prominent and technically sophisticated cashier-less retail system in existence (Demaitre, 2018). First opened to the public in Seattle, Washington in January 2018, Amazon Go stores use a combination of computer vision, deep learning, and sensor fusion to enable a fully automated shopping experience under the brand's slogan: 'Just Walk Out Technology'.

Figure 2.1 — Amazon Go Store: Just Walk Out Technology, Seattle USA (2018) Source: Amazon.com, Inc. (2018). Amazon Go Store Launch. amazon.com/go [Accessed April 2025]

2.2.1 System Architecture and Technology

The Amazon Go system relies on a dense array of overhead cameras mounted throughout the store, capable of tracking customer movements and identifying which products are picked from or returned to shelves (Florentine, 2018). This tracking is supported by shelf-mounted weight sensors that detect subtle changes when items are removed, providing an additional layer of confirmation through sensor fusion. Computer vision algorithms, trained on massive datasets using deep convolutional neural networks, associate customers with a virtual cart throughout their visit.

2.2.2 Strengths

The principal strength of Amazon Go is the completeness of its automation — no scanning, no queuing, no manual payment initiation (Demaitre, 2018). The high accuracy of AI tracking minimizes billing errors and inventory discrepancies, and the system demonstrated for the first time at commercial scale that fully autonomous retail is technically feasible.

2.2.3 Limitations and Relevance to SCAN_GO

Despite its achievements, Amazon Go suffers from critical scalability and cost limitations. The installation cost for a single store has been estimated at several million US dollars, requiring hundreds of cameras, proprietary sensor hardware, and significant computational infrastructure (Hamilton, 2019). It also requires major store layout modifications, making it impractical for retrofitting existing supermarkets. SCAN_GO directly addresses this barrier by replacing store-wide overhead infrastructure with a smart basket-integrated barcode scanner and weight sensor, eliminating the need for any structural store modifications.

Key Takeaway from Amazon Go:

While Amazon Go proves the viability of cashier-less retail at a premium cost, SCAN_GO achieves a comparable outcome — eliminating checkout queues and enabling digital payment — through a basket-first, infrastructure-light approach that is orders of magnitude cheaper to implement.

2.3 Standard Cognition — AI-Powered Autonomous Checkout (USA)

Standard Cognition is a San Francisco-based startup founded in 2017, positioning itself as a provider of autonomous checkout technology deployable in existing retail stores without requiring a complete overhaul (Standard Cognition, 2021). The company uses ceiling-mounted AI cameras and computer vision software to track customers and their product selections.

Figure 2.2 — Standard Cognition: AI Camera-Based Autonomous Checkout Platform Source: Standard Cognition Corp. (2021). Autonomous Checkout Technology. standard.ai [Accessed April 2025]

2.3.1 System Description and Closure

The company raised over USD 150 million in venture funding and partnered with retailers in North America and Japan. However, in January 2023, Standard Cognition announced it was shutting down, citing the extreme difficulty and cost of achieving reliable autonomous checkout at commercial scale (Perez, 2023). This closure is a critical data point demonstrating that even well-funded vision-based systems face fundamental viability challenges.

2.3.2 Key Lessons for SCAN_GO

For SCAN_GO, this outcome reinforces the decision to rely on explicit hardware-integrated barcode scanning mounted on the shopping basket rather than passive computer vision tracking. By making product identification a deliberate, basket-level hardware action, SCAN_GO eliminates store-wide infrastructure investment while achieving cashier-less checkout.

2.4 AiFi — Scalable Autonomous Retail Platform (USA / Europe)

AiFi is a prominent player in autonomous retail, founded in 2018 in Santa Clara, California. Unlike Amazon Go, AiFi's platform is explicitly designed for multi-retailer deployment as a white-label solution (AiFi, 2022). The system uses overhead cameras combined with deep learning for real-time shopper and product tracking.

Figure 2.3 — AiFi: Multi-Retailer Autonomous Checkout Platform (USA/Europe) Source: AiFi Inc. (2022). Autonomous Retail Solutions. aifi.io [Accessed April 2025]

2.4.1 Deployments

AiFi has deployed its technology in collaboration with various retail chains in the United States and Netherlands, including a partnership with OAK in Amsterdam and stadium concessions at State Farm Arena (AiFi, 2022). The system scales from small convenience stores to larger supermarket formats.

2.4.2 Comparison with SCAN_GO

AiFi represents an improvement over Amazon Go in business model flexibility, but underlying technology costs remain high. SCAN_GO requires no camera installation whatsoever; product identification is handled by a dedicated hardware barcode scanner embedded in each smart basket — inherently more scalable to budget-constrained environments such as Egyptian supermarkets.

2.5 European Smart Retail Systems

Europe has been a significant arena for smart retail development, with major grocery chains investing in approaches ranging from AI-powered autonomous stores to mobile-app-integrated checkout systems. This section examines the most prominent European implementations.

2.5.1 Tesco GetGo — United Kingdom

Tesco, the UK's largest supermarket chain, launched its GetGo cashier-less format in London in 2021 in partnership with Trigo Vision, an Israeli startup specializing in frictionless retail (Butler, 2021). The store uses overhead cameras and deep learning computer vision to automatically track products customers pick up, charging them through the Tesco app.

Figure 2.4 — Tesco GetGo: UK's First Major Cashierless Supermarket (London, 2021) Source: Tesco PLC / Trigo Vision (2021). GetGo Store Launch. tescoplc.com [Accessed April 2025]

Compared to SCAN_GO, Tesco GetGo requires dedicated ceiling camera installations and significant backend computing investment (Butler, 2021). It is also designed for small-format convenience stores, limiting scalability. SCAN_GO, using a basket-integrated scanner and weight sensor, requires no store-side hardware modifications beyond the smart baskets themselves.

2.5.2 Rewe Pick&Go — Germany

Rewe launched its Pick&Go autonomous checkout concept in Cologne in 2021, using overhead cameras and AI-based product recognition with QR-code customer authentication via the Rewe mobile app (Retail Technology Innovation Hub, 2022). Early pilot results indicated high customer satisfaction with average shopping trips under 15 minutes.

Figure 2.5 — Rewe Pick&Go: Germany's Autonomous Supermarket Concept (Cologne, 2021) Source: Rewe Group (2022). Pick&Go Autonomous Store. rewe-group.com [Accessed April 2025]

2.5.3 Carrefour Flash — France

Carrefour opened its first autonomous store, Carrefour Flash, in Paris in 2019 in collaboration with AiFi (Green Seed Group, 2019). The store demonstrates that major traditional supermarket chains can adapt to new technology paradigms, but has remained limited to small-format locations due to high per-store infrastructure costs — a limitation SCAN_GO directly avoids.

Figure 2.6 — Carrefour Flash: France's Fully Automated Store (Paris, 2019) Source: Carrefour Group (2019). Carrefour Flash Store. carrefour.com [Accessed April 2025]

2.5.4 Summary of European Approaches

European smart retail initiatives share common characteristics: computer vision and AI for autonomous product identification, significant store infrastructure investment, primarily small-format deployment, and mobile app integration for authentication. While achieving genuine frictionless checkout, their cost and complexity limit deployment to well-funded retailers in developed markets. SCAN_GO draws inspiration from these customer experience goals while addressing their fundamental cost limitation.

2.6 Asian Smart Retail Systems

Asia, particularly China, Japan, and South Korea, has been at the forefront of smart retail innovation. Several Asian systems prioritize mobile-based solutions and QR code technology, aligning more closely with SCAN_GO's design philosophy than their Western counterparts.

2.6.1 Alibaba Freshippo (Hema) — China

Alibaba's Freshippo (Hema) represents one of the most ambitious smart retail implementations globally. Launched in 2016, it integrates online and offline shopping through the Alipay mobile app, combining RFID tags, barcode scanning via smartphones, and face recognition technology at checkout (Liu & Xu, 2020). Freshippo demonstrates that a barcode-scanning approach can scale to large supermarket formats.

Figure 2.7 — Alibaba Freshippo (Hema): China's New Retail Experience (Launched 2016) Source: Alibaba Group (2020). Freshippo New Retail. alibabagroup.com [Accessed April 2025]

However, Freshippo's reliance on face recognition raises significant privacy concerns and requires expensive biometric hardware (Liu & Xu, 2020). SCAN_GO replaces face recognition with QR code-based exit verification, achieving similar security outcomes without privacy implications or hardware costs beyond the smart basket unit itself.

2.6.2 JD.com X-Mart — China

JD.com launched its X-Mart concept stores in Beijing and Shanghai in 2018, featuring AI cameras, RFID-tagged products, and autonomous robotic restocking systems (JD.com, 2018). While technologically impressive, X-Mart represents a maximum-investment solution — SCAN_GO achieves cashier-less checkout using a basket-embedded barcode scanner and weight sensor, with no RFID tags, overhead cameras, or robots required.

Figure 2.8 — JD.com X-Mart: AI + RFID + Robots Future Store (China, 2018) Source: JD.com Inc. (2018). X-Mart Future Store Concept. jd.com [Accessed April 2025]

2.6.3 BingoBox — China

BingoBox pioneered the fully unmanned, small-format convenience store, founded in 2016. Stores use RFID readers, QR code scanning for entry, and self-checkout kiosks with WeChat Pay/Alipay integration (TechNode, 2018).

BingoBox is perhaps the closest Chinese comparison for SCAN_GO because both rely on QR code-based verification and digital payment. However, BingoBox's RFID approach requires all products to be individually tagged — SCAN_GO uses standard barcodes already printed on products, read by the basket-integrated hardware scanner, eliminating product-level hardware modification entirely.

Figure 2.9 — BingoBox: China's Unmanned Convenience Store with RFID Technology Source: BingoBox Retail Technology (2018). Unmanned Store Concept. bingobox.com [Accessed April 2025]

2.6.4 Smart Retail in South Korea

South Korea has been a regional leader in retail digitization. GS25 and Lotte have piloted cashier-less formats using IoT shelf sensors and QR code-based digital payment, reflecting the country's high smartphone penetration and digital payment adoption (Korea Retail Institute, 2021). Korea's experience demonstrates that QR code and barcode-based checkout can achieve widespread adoption — evidence that SCAN_GO's foundational approach is commercially viable.

Figure 2.10 — Smart Retail Korea: IoT + QR Code + Mobile Payment Integration Source: Korea Retail Institute (2021). Smart Store Technology Report. retailkorea.or.kr [Accessed April 2025]

2.6.5 Seven-Eleven Japan — Cashierless Pilots

Seven-Eleven Japan has conducted extensive cashier-less pilots in partnership with AiFi and NEC, driven by Japan's acute retail labor shortage due to an aging population (NEC Corporation, 2020). Japanese pilots combine basket-level scanning, AI cameras, and IC card-based payment. Notably, Japan's experience highlights that basket-based checkout must be designed with accessibility in mind — directly informing SCAN_GO's interface design decisions prioritizing simplicity for all age groups.

Figure 2.11 — Seven-Eleven Japan: Smart Cashierless Store Technology Pilot Source: NEC Corporation (2020). Seven-Eleven Japan Unmanned Store Pilot. nec.com [Accessed April 2025]

2.7 Review of Academic Literature

Beyond commercial implementations, substantial academic research has investigated the technical challenges, human factors, and business implications of smart retail and cashier-less checkout systems.

2.7.1 Mobile Self-Scanning and Consumer Behavior

Research by Hossain et al. (2020) examined adoption of self-scanning applications in European supermarkets, finding that perceived ease of use, time savings, and privacy concerns were the primary factors influencing adoption intention. Customers using self-scanning systems reported significantly higher satisfaction with the checkout experience, primarily due to reduced waiting times.

A complementary study by Van der Heijden and Sorber (2019) found that providing clear visual feedback on cart contents and total cost during shopping significantly increases perceived usefulness and actual usage frequency — a finding directly incorporated into SCAN_GO's Android tablet interface through real-time cart display and running total calculation.

2.7.2 Barcode Scanning Accuracy and Performance

Müller and Richter (2021) evaluated barcode scanning accuracy under retail lighting conditions. Testing EAN-13 and UPC-A barcodes across fifteen scanner configurations, they found modern hardware barcode scanners achieved accuracy rates exceeding 98% under standard supermarket lighting, rising to over 99.5% in well-lit environments — confirming SCAN_GO's technical viability for commercial deployment.

2.7.3 QR Code-Based Verification Systems

Chen et al. (2022) analyzed QR-based payment verification in Chinese digital payment systems. Time-limited QR codes with cryptographic signing mechanisms provided robust security against fraud while maintaining checkout verification under three seconds on average. SCAN_GO's exit verification mechanism draws directly on this research model.

2.7.4 Real-Time Inventory Management

Garaus and Wagner (2019) analyzed inventory management impact of smart checkout systems across fifteen supermarket pilots in Austria and Germany, finding real-time transaction recording reduced inventory discrepancies by an average of 34% compared to traditional batch updates. This directly supports SCAN_GO's inclusion of a real-time administrative dashboard as a core system component.

2.7.5 Digital Payment Adoption in Developing Markets

Ozili (2018) found that acceptance of digital payment solutions in African markets was strongly correlated with perceived security, ease of use, and alignment with existing payment habits. Egypt has experienced significant growth in digital payment adoption, with the Central Bank of Egypt reporting a 300% increase in mobile wallet transactions between 2019 and 2023 — validating SCAN_GO's payment integration strategy for the Egyptian market.

2.8 Traditional Self-Checkout Kiosks — A Transitional Technology

Research by Collier et al. (2021) found that self-checkout kiosks reduce cashier labor costs by 40% on average while improving throughput by 25% during peak hours. However, they still require customers to checkout at a fixed terminal, meaning queues persist. The ECR Retail Loss Group (2020) further found kiosks were associated with shrinkage rates 2–3 times higher than cashier lanes, due to intentional and accidental scanning failures.

SCAN_GO addresses both limitations: by distributing scanning throughout the shopping journey via the basket-integrated hardware scanner and weight sensor, checkout queues are eliminated entirely; and the QR code exit verification gate ensures only customers who completed payment can exit — providing stronger anti-theft assurance than kiosk-only implementations.

2.9 Comprehensive Comparative Analysis

The following table provides a structured comparison of SCAN_GO against the principal smart retail systems reviewed in this chapter. Cost and scalability ratings are defined in the footnotes below the table.

System	Country	Core Technology	HW Cost (a)	Scalability (b)	Cashierless	Basket/App	SCAN_GO Advantage
Amazon Go	USA	Computer Vision + AI + Sensor Fusion	Very High	Low	✓Yes	Partial	Much lower cost; no infra needed
Standard Cognition	USA	AI Camera Networks (Ceiling-mounted)	Very High	Low	✓Yes	✗No	Avoids heavy AI infra; still operational
AiFi	USA/EU	Autonomous AI Checkout (White-label)	High	Medium	✓Yes	Partial	Open to any retailer; no proprietary lock-in
Tesco GetGo	UK	Computer Vision + Deep Learning (Trigo)	High	Low	✓Yes	✓Yes	No ceiling camera installation required

System	Country	Core Technology	HW Cost (a)	Scalability (b)	Cashierless	Basket/App	SCAN_GO Advantage
Rewe Pick&Go	Germany	Overhead Cameras + AI Tracking	High	Low	✓Yes	✓Yes	No fixed infrastructure modifications
Carrefour Flash	France	Computer Vision Cameras (AiFi-powered)	High	Low	✓Yes	✓Yes	Cost-effective alternative approach
Alibaba Freshippo	China	RFID + Face Recognition + AI	High	Medium	✓Yes	✓Yes	No face recognition; preserves privacy
JD.com X-Mart	China	AI Cameras + RFID Tags + Autonomous Robots	Very High	Low	✓Yes	Partial	No RFID tagging or robotic systems needed
BingoBox	China	RFID + QR Entry + Self-Checkout Kiosk	Medium	High	Partial	✓Yes	Fully basket-based; no RFID product tagging
Smart Retail Korea	S. Korea	IoT Shelf Sensors + QR + Digital Payment	Medium	High	Partial	✓Yes	Simpler deployment; lower per-store cost
SCAN_GO (Proposed)	Egypt	Basket Scanner + Weight Sensor + ESP32 + QR Exit Verify	Very Low	Very High	✓Yes	✓Yes	— Proposed System —

Table 2.1 — Comparative Analysis of Smart Retail and Cashier-less Checkout Systems

Table Notes & Definitions:

(a) Hardware Cost Scale: Very High = >$1M per store installation | High = $200K–$1M | Medium = $20K–$200K | Very Low = <$5K (basket hardware per unit only, no store modifications)

(b) Scalability Scale: Very High = deployable across hundreds of stores with minimal incremental cost | High = moderate hardware per store | Low = requires store-specific installation (cameras, sensors)

(c) Cashier-less: ✓Yes = no cashier required at any point | Partial = reduced cashier role (e.g. kiosk assistance still needed) | ✗No = cashier still required

Sources: Demaitre (2018); Standard Cognition (2021); AiFi (2022); Butler (2021); Retail Technology Innovation Hub (2022); Green Seed Group (2019); Liu & Xu (2020); JD.com (2018); TechNode (2018); Korea Retail Institute (2021).

The comparative analysis reveals that all high-profile cashier-less implementations rely on computer vision and overhead camera infrastructure, imposing high costs limiting applicability to well-funded retailers in developed markets. Systems most closely approaching SCAN_GO's cost profile — BingoBox and Korean IoT/QR systems — still rely on hardware modifications at the product or shelf level. SCAN_GO uniquely achieves cashier-less checkout by embedding intelligence directly into the shopping basket: a dedicated barcode scanner, an HX711-driven weight verification module, and an Android tablet interface — requiring no per-product tagging, no sensor installation in the store, and no camera systems.

2.10 Identification of the Research Gap

The literature review reveals a clear research gap: the absence of a cost-effective, basket-integrated, infrastructure-light cashier-less checkout solution suitable for deployment in supermarkets in developing and emerging market contexts (Ozili, 2018; Hossain et al., 2020).

Existing solutions fall into two categories. The first — autonomous computer vision systems — achieves genuine cashier-less checkout at prohibitive infrastructure costs. The second — kiosks and RFID systems — reduces cashier reliance but requires hardware investment and does not fully eliminate checkout queues.

Basket-integrated self-scanning systems have been explored in academic research (Van der Heijden & Sorber, 2019) and limited commercial deployments, but these lack integrated exit verification mechanisms and have not been designed for the economic constraints of developing market retail environments. SCAN_GO fills this gap with a comprehensive architecture — basket-embedded barcode scanning, real-time weight verification via the HX711 load cell, real-time cart management on the Android tablet, integrated digital payment, exit QR verification gate, and administrative inventory monitoring — that achieves full cashier-less capabilities without any store-side infrastructure modification beyond deploying the smart baskets.

2.11 Chapter Summary

This chapter has provided a comprehensive review of the existing landscape of smart retail and cashier-less checkout systems, examining implementations from the United States, United Kingdom, Germany, France, China, South Korea, and Japan, alongside relevant academic research on self-scanning, barcode accuracy, QR code verification, inventory management, and digital payment adoption.

The review demonstrates that while significant technological progress has been made, existing solutions are characterized by high implementation costs, dependency on custom infrastructure, and limited applicability to retail environments with constrained capital budgets. Traditional self-checkout kiosks fail to eliminate checkout queues and present unresolved theft prevention challenges.

Academic research confirms the technical viability and consumer acceptance of hardware barcode scanning as a checkout mechanism, and supports the QR code verification approach adopted by SCAN_GO for exit security. Research on digital payment adoption in developing markets provides evidence that Egyptian consumers are increasingly prepared to engage with digital payment solutions.

Together, these findings establish a clear research gap and validate SCAN_GO's proposed basket-integrated approach as a novel and practically viable contribution. The following chapter will present the system design and architecture of SCAN_GO.

References

- AiFi Inc. (2022). Autonomous retail solutions overview. Retrieved from https://aifi.io
- Butler, S. (2021). Tesco opens first cashierless store in UK. The Guardian. guardian.com
- Central Bank of Egypt. (2023). Annual Report: Digital Payments Growth 2019–2023. Cairo: CBE.
- Chen, L., Wang, Y., & Zhang, H. (2022). QR code-based payment verification systems: Security and performance analysis. Electronic Commerce Research and Applications, 51, 101–119.
- Collier, J. E., Moore, R. S., Horky, A., & Moore, M. L. (2021). Why the interaction matters: Retail associate and self-checkout redemption. Journal of Retailing, 97(1), 42–58.
- Demaitre, E. (2018). Amazon Go: The cashierless convenience store opens to the public. The Robot Report. therobotreport.com
- ECR Retail Loss Group. (2020). Self-checkout shrinkage study: European retail perspectives. Brussels: ECR Community.
- Florentine, S. (2018). How Amazon Go works: A visual explanation. CIO Magazine.
- Garaus, M., & Wagner, U. (2019). Smart retail technology and inventory management. Journal of Retailing, 95(4), 401–418.
- Green Seed Group. (2019). Carrefour opens Flash, the fully automated store. greenseedgroup.com

- Grewal, D., Roggeveen, A. L., & Nordfält, J. (2017). The future of retailing. Journal of Retailing, 93(1), 1–6.
- Hamilton, I. A. (2019). Amazon Go stores cost millions to build — here's what goes into them. Business Insider.
- Hossain, M., Sinha, A., Bashir, M., & Jain, N. (2020). Mobile self-scanning in supermarkets: Drivers of adoption. Journal of Retailing and Consumer Services, 52, 101–113.
- JD.com Inc. (2018). JD X-Mart: The future store concept. Retrieved from https://jd.com
- Korea Retail Institute. (2021). Smart store technology and digital checkout adoption in South Korea. Seoul: KRI.
- Liu, Z., & Xu, J. (2020). New retail in China: Alibaba Freshippo and the integration of online and offline. Journal of Retailing and Consumer Services, 55, 102–118.
- Müller, T., & Richter, K. (2021). Barcode scanning accuracy under retail conditions. International Journal of Retail & Distribution Management, 49(3), 321–337.
- NEC Corporation. (2020). Seven-Eleven Japan unmanned store pilot results. Retrieved from https://nec.com
- Ozili, P. K. (2018). Impact of digital finance on financial inclusion and stability. Borsa Istanbul Review, 18(4), 329–340.
- Pantano, E., Pizzi, G., Scarpi, D., & Dennis, C. (2018). Competing through digitization: Understanding the role of digital-only retailing. Journal of Retailing and Consumer Services, 44, 8–15.
- Perez, S. (2023). Standard Cognition, the autonomous checkout startup, shuts down. TechCrunch. techcrunch.com
- Retail Technology Innovation Hub. (2022). Rewe Pick&Go: Sixth autonomous checkout store. retailtechinnovationhub.com
- Standard Cognition Corp. (2021). Autonomous checkout technology overview. Retrieved from https://standard.ai
- TechNode. (2018). BingoBox unveils new AI tech for its unmanned stores. technode.com
- Van der Heijden, H., & Sorber, E. (2019). Consumer acceptance of mobile shopping assistants in supermarkets. Computers in Human Behavior, 96, 1–12.